

Vinayak Nandikole

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Summary

Full-Stack Engineer (React, Node.js, MongoDB) shipping production web platforms serving up to **50K applicants/year** at Teach For India. Specialty: turning tangled enterprise workflows (**Salesforce**, OAuth, **Algolia**) into clean user-facing flows that non-technical teams can actually operate.

Skills

Languages	JavaScript, TypeScript, Java, Python, C++, SQL
Frontend	React (17/18), Redux, Material UI, React Router, Leaflet, HTML, CSS
Backend & Data	Node.js, Express, Mongoose, Bull (queues), MongoDB, Redis, REST APIs, PL/SQL
Cloud & Integrations	AWS S3, Algolia, Salesforce (jsforce, REST/Bulk API), Passport.js OAuth (Google/LinkedIn), PM2, Whisper API
Tools & DevOps	Git, GitHub, Docker, CI/CD, PM2 Cluster Mode, Claude Code, Antigravity
Certifications	AWS Academy Cloud Foundations, Microsoft Azure Fundamentals (AZ-900)

Work Experience

Teach For India | Associate, Web Development

Aug 2023 – Present, Pune, India

- Owned full-stack development across three flagship portals: **Fellowship** (primary, up to **50K applicants/year**), **Volunteer**, and **Internship**; led the **Volunteer Portal revamp** end-to-end from UX design through production deployment.
- Architected **async job processing** with **Bull + Redis** across 20+ job types (Salesforce syncs, assessment scoring, data synchronization), running on dedicated **PM2-clustered worker processes** separated from the API for peak-cycle scalability.
- Enhanced internal **Kitchen** admin tooling: automated Salesforce interview slot fetch with real-time capacity deduction and built a **mock application system** that exercises the full applicant waterfall for regression testing.
- Integrated **Salesforce** bidirectionally across 8+ objects with multi-provider auth (Google/LinkedIn OAuth, OTP) via **Passport.js**; automated DB operations cutting error rate by **15%** and resolved 50+ production bugs across a 4-engineer team.
- Designed a feature-flag system (e.g., `allowACOperations`) governing runtime behavior of picklists and dashboard actions, enabling safe rollouts of operational changes without code deploys.

Hexaware Technologies | Associate Software Engineer

June 2022 – Apr 2023, Pune, India

- Migrated **PL/SQL** packages to **SQL** for the **Oracle to Snowflake** data warehouse transition, ensuring zero data loss across production schemas.
- Collaborated in a cross-functional Git/GitHub workflow with code reviews, issue tracking, and release cadences on a multi-engineer data modernization team.

Selected Projects

Fellowship Application Portal

apply.teachforindia.org

- Architected a **monorepo** (Express server, React 18 applicant portal, admin Kitchen dashboard) with **cohort-scoped MongoDB collections** for year-over-year data isolation; cut load time by **20%** via code splitting and caching.
- Built **video proctoring** for in-house assessments using **COCO-based person detection** and **face-embedding recognition** (cosine similarity) with pre-flight checks, violation reporting, and S3 evidence snapshots.
- Engineered video recorder for interview responses (browser-specific guidance, WebM fixes, S3 upload, **Whisper transcription**) and **dynamic PDF generation** for offer letters and CVs.

Volunteer Management System

volunteer.teachforindia.org

- Re-architected with **action-first UX**: replaced dropdowns with chips/cards for cities, clusters, and subjects; added global search with auto-suggestions and step-wise auto-scroll, cutting clicks by **36%**.
- Built **Algolia-Salesforce sync microservice** with smart update logic that auto-removes fulfilled, inactive, or referred opportunities from the index via queue-based processing.
- Unified volunteer and internship opportunities behind a single Algolia `recordType` filter, eliminating the need for separate indices and simplifying operational upkeep.

Impacted Internship Program

impacted.teachforindia.org

- Launched as a sibling CRA app with **referral-gated access** via unique reference keys for invite-only onboarding and a pre-filled application flow to reduce drop-off.
- Designed a backend classification toggle distinguishing internship vs volunteer applicants, sharing infrastructure with the Volunteer Portal for zero-duplication data sync.

Achievements

4-Star in C++ at HackerRank

[Profile](#)

Ranked **GeeksforGeeks institutional rank 15** with **130+ problems solved** across DSA, algorithms, and competitive programming domains.

150+ LeetCode Problems Solved

[Profile](#)

Consistent focus on data structures, algorithms, and system design problem solving.

Education

Marathwada Mitra Mandal's College of Engineering

B.E. in Electrical Engineering - CGPA: 9.02/10

June 2022
Pune, Maharashtra